EXPERIMENT-09

Working with Intents. Create an android application that uses both implicit and explicit intents to navigate between different activities. Implement intent filters for specific actions.

Aim: To create an Android application that uses Explicit and Implicit Intents to navigate between activities and implement Intent Filters.

Theory:

1. **What is an Intent?**
   An **Intent** is a messaging object used to request an action from another component in Android.
   Intents are mainly used for:
   - Starting another activity
   - Opening system applications
   - Passing data between activities

2. **Types of Intents**
   I. Explicit Intent
   Explicit intent specifies the **exact activity** to start.
   Example:
   - Move from **MainActivity → SecondActivity**

   II. Implicit Intent
   Implicit intent **does not specify the target activity**. Android finds a suitable application to perform the action.
   Example:
   - Open browser
   - Open dialer
   - Share text

3. **Project Structure**
   - MainActivity
   - SecondActivity
   - activity_main.xml
   - activity_second.xml
   - AndroidManifest.xml

Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center">
        <Button
            android:id="@+id/btnExplicit"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Open Second Activity"/>
        <Button
            android:id="@+id/btnBrowser"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Open Website"
            android:layout_marginTop="20dp"/>
        <Button
            android:id="@+id/btnDial"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Open Dialer"
            android:layout_marginTop="20dp"/>
    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.experiment_09;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.Button;
import android.content.Intent;
import android.net.Uri;

public class MainActivity extends AppCompatActivity {
    2 usages
    Button btnExplicit, btnBrowser, btnDial;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnExplicit = findViewById(R.id.btnExplicit);
        btnBrowser = findViewById(R.id.btnBrowser);
        btnDial = findViewById(R.id.btnDial);
// Explicit Intent
        btnExplicit.setOnClickListener( View v -> {
            Intent intent = new Intent( packageContext: MainActivity.this, SecondActivity.class);
            startActivity(intent);
        });
// Implicit Intent - Open Website
        btnBrowser.setOnClickListener( View v -> {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse( uriString: "https://www.google.com"));
            startActivity(intent);
        });
// Implicit Intent - Open Dialer
        btnDial.setOnClickListener( View v -> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse( uriString: "tel:1234567890"));
            startActivity(intent);
        });
    }
}
```

activity_second.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SecondActivity">

    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Welcome to Second Activity"
            android:textSize="22sp"/>
    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

SecondActivity.java

```java
package com.example.experiment_09;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

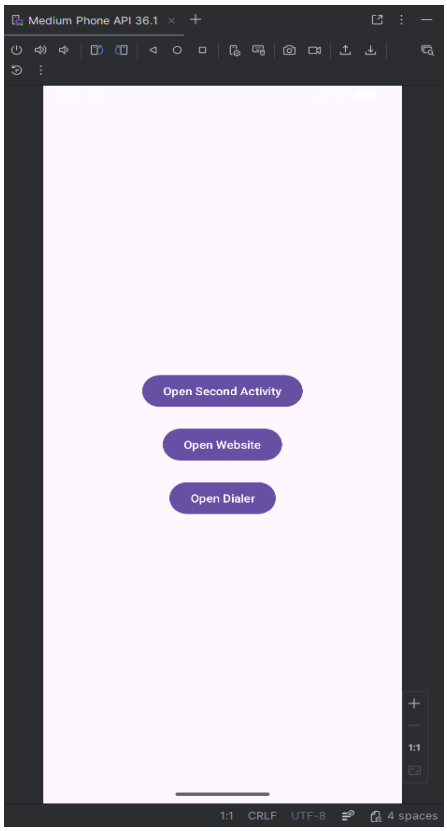
AndroidManifest.xml (Intent Filter)

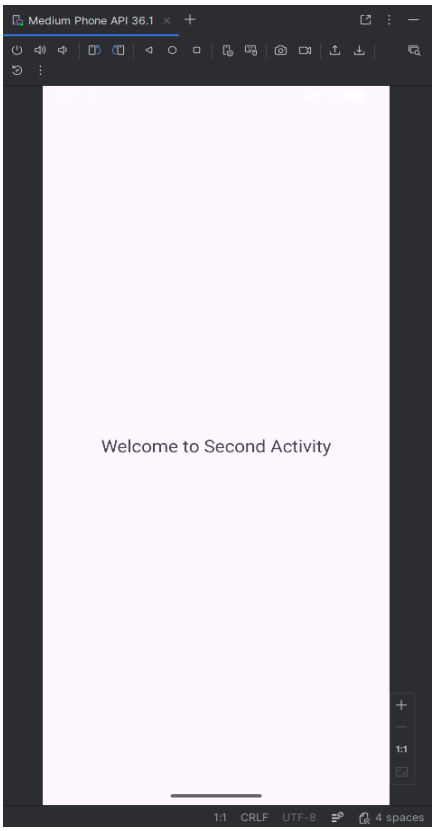
```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="Experiment-09"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Experiment09">

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".SecondActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.VIEW"/>
                <category android:name="android.intent.category.DEFAULT"/>
                <category android:name="android.intent.category.BROWSABLE"/>
                <data android:scheme="https"
                    android:host="www.google.com"/>
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Output:

1) Explicit Intent

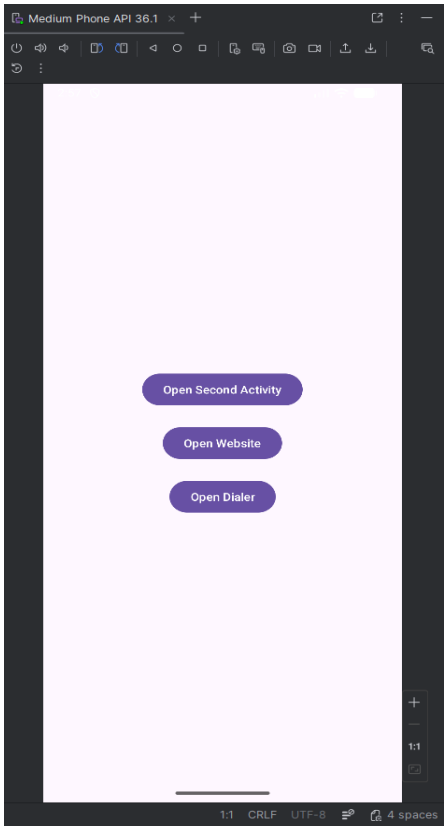
Button click: **MainActivity → SecondActivity**

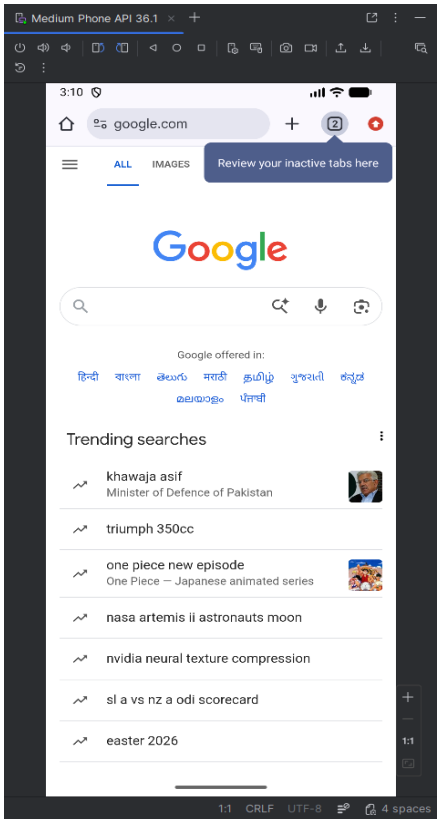




2) Implicit Intent (Browser)

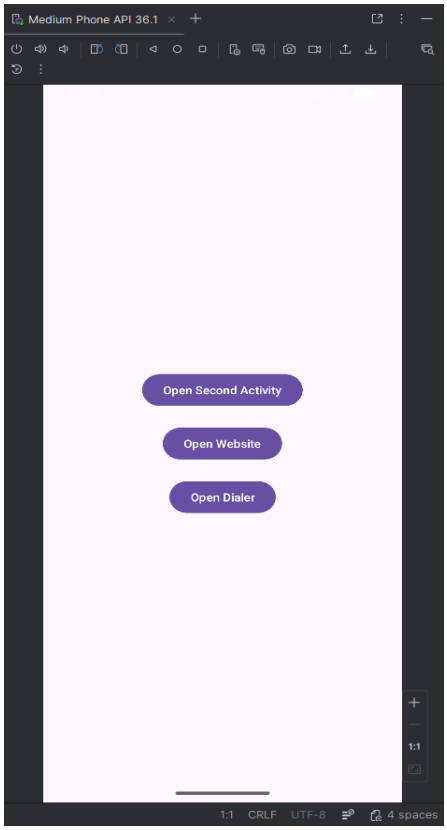
Button click: **Open Website → Browser opens google.com**

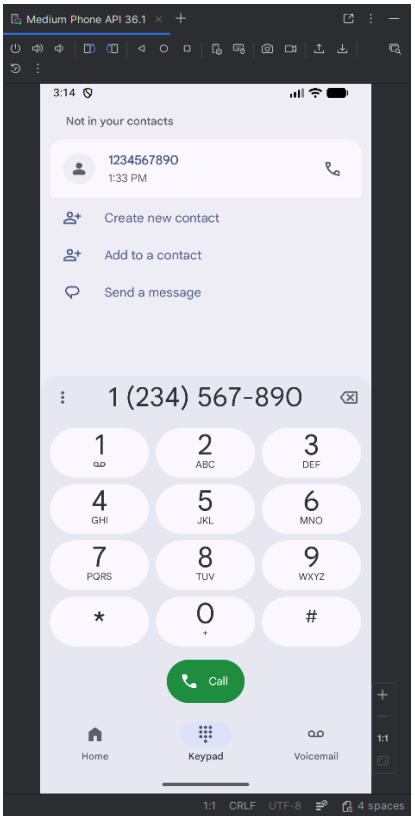




3) Implicit Intent (Dialer)

Button click: **Dialer opens with number 1234567890**





Conclusion:

This experiment demonstrated how Explicit Intents can be used to navigate between activities within the same application, while Implicit Intents enable interaction with external apps based on user actions. It also helped in understanding the role of Intent Filters in defining how activities respond to specific actions. Overall, the practical enhanced knowledge of Android component communication and activity management.